COMPSCI 402 Assignment 4

Q1. RL 2-point

Pacman is in an unknown MDP where there are three states [A, B, C] and two actions [Stop, Go]. We are given the following samples generated from taking actions in the unknown MDP. For the following problems, assume $\gamma = 1$ and $\alpha = 0.5$.

(a) We run Q-learning on the following samples: 1-point

s	a	s'	r
A	Go	B	2
C	Stop	A	0
B	Stop	A	-2
B	Go	C	-6
C	Go	A	2
A	Go	A	-2

What are the estimates for the following Q-values as obtained by Q-learning? All Q-values are initialized to 0.

(i) $Q(C, Stop) =$ _____

(ii) $Q(C, Go) =$ _____

(b) For this next part, we will switch to a feature based representation. We will use two features: 1-point

- $f_1(s, a) = 1$
- $f_2(s, a) = \begin{cases} 1 & a = \text{Go} \\ -1 & a = \text{Stop} \end{cases}$

Starting from initial weights of 0, compute the updated weights after observing the following samples:

s	a	s'	r
A	Go	B	4
B	Stop	A	0

What are the weights after the first update? (using the first sample)

(i) $w_1 =$ _____

(ii) $w_2 =$ _____

What are the weights after the second update? (using the second sample)

(iii) $w_1 =$ _____

(iv) $w_2 =$ _____

Q2. Q-learning 4-point

Consider an unknown MDP with three states (A, B and C) and two actions ($\leftarrow$ and $\rightarrow$). Suppose the agent chooses actions according to some policy π in the unknown MDP, collecting a dataset consisting of samples (s, a, s', r) representing taking action a in state s resulting in a transition to state s' and a reward of r.

s	a	s'	r
A	$\rightarrow$	B	2
C	$\leftarrow$	B	2
B	$\rightarrow$	C	-2
A	$\rightarrow$	B	4

You may assume a discount factor of $\gamma = 1$.

(a) Recall the update function of Q-learning is:

$$Q(s_t, a_t) \leftarrow (1-\alpha)Q(s_t, a_t) + \alpha\left(r_t + \gamma \max_{a'} Q(s_{t+1}, a')\right)$$

Assume that all Q-values are initialized to 0, and use a learning rate of $\alpha = \frac{1}{2}$.

(i) Run Q-learning on the above experience table and fill in the following Q-values: 1-point

$Q(A, \rightarrow) =$ _____ $Q(B, \rightarrow) =$ _____

(ii) After running Q-learning and producing the above Q-values, you construct a policy π_Q that maximizes the Q-value in a given state:

$$\pi_Q(s) = \arg\max_a Q(s, a).$$

What are the actions chosen by the policy in states A and B? 1-point

$\pi_Q(A)$ is equal to:

- ◯ $\pi_Q(A) = \leftarrow$.
- ◯ $\pi_Q(A) = \rightarrow$.
- ◯ $\pi_Q(A) =$ Undefined.

$\pi_Q(B)$ is equal to:

- ◯ $\pi_Q(B) = \leftarrow$.
- ◯ $\pi_Q(B) = \rightarrow$.
- ◯ $\pi_Q(B) =$ Undefined.

(b) Use the empirical frequency count model-based reinforcement learning method described in lectures to estimate the transition function $\hat{T}(s, a, s')$ and reward function $\hat{R}(s, a, s')$. (Do not use pseudocounts; if a transition is not observed, it has a count of 0.)

Write down the following quantities. You may write N/A for undefined quantities. 1-point

$\hat{T}(A, \rightarrow, B) =$ _____ $\hat{R}(A, \rightarrow, B) =$ _____

$\hat{T}(B, \rightarrow, A) =$ _____ $\hat{R}(B, \rightarrow, A) =$ _____

$\hat{T}(B, \leftarrow, A) =$ _____ $\hat{R}(B, \leftarrow, A) =$ _____

(c) This question considers properties of reinforcement learning algorithms for *arbitrary* discrete MDPs; you do not need to refer to the MDP considered in the previous parts. 1-point

(i) Which of the following methods, at convergence, provide enough information to obtain an optimal policy? (Assume adequate exploration.)

- ☐ Model-based learning of T(s,a,s') and R(s,a,s').
- ☐ Direct Evaluation to estimate V(s).
- ☐ Temporal Difference Learning to estimate V(s).
- ☐ Q-Learning to estimate Q(s,a).

(ii) In the limit of infinite timesteps, under which of the following exploration policies is Q-learning guaranteed to converge to the optimal Q-values for all state? (You may assume the learning rate α is chosen appropriately, and that the MDP is ergodic: i.e., every state is reachable from every other state with non-zero probability.)

- ☐ A fixed policy taking actions uniformly at random.
- ☐ A greedy policy.
- ☐ An ϵ-greedy policy
- ☐ A fixed optimal policy.

Q3. Reinforcement Learning 2-point

Imagine an unknown environments with four states (A, B, C, and X), two actions ($\leftarrow$ and $\rightarrow$). An agent acting in this environment has recorded the following episode:

s	a	s'	r	Q-learning iteration numbers (for part b)
A	$\rightarrow$	B	0	1, 10, 19, ...
B	$\rightarrow$	C	0	2, 11, 20, ...
C	$\leftarrow$	B	0	3, 12, 21, ...
B	$\leftarrow$	A	0	4, 13, 22, ...
A	$\rightarrow$	B	0	5, 14, 23, ...
B	$\rightarrow$	A	0	6, 15, 24, ...
A	$\rightarrow$	B	0	7, 16, 25, ...
B	$\rightarrow$	C	0	8, 17, 26, ...
C	$\rightarrow$	X	1	9, 18, 27, ...

(a) Consider running model-based reinforcement learning based on the episode above. Calculate the following quantities: 0.5-point

$\hat{T}(B, \rightarrow, C) =$ _____

$\hat{R}(C, \rightarrow, X) =$ _____

(b) Now consider running Q-learning, repeating the above series of transitions in an infinite sequence. Each transition is seen at multiple iterations of Q-learning, with iteration numbers shown in the table above.

After which iteration of Q-learning do the following quantities first become nonzero? (If they always remain zero, write *never*). 0.5-point

$Q(A, \rightarrow)$? _____

$Q(B, \leftarrow)$? _____

(c) True/False: For each question, you will get positive points for correct answers, zero for blanks, and negative points for incorrect answers. Circle your answer **clearly**, or it will be considered incorrect. 1-point

(i) [*true* or *false*] In Q-learning, you do not learn the model.

(ii) [*true* or *false*] For TD Learning, if I multiply all the rewards in my update by some nonzero scalar p, the algorithm is still guaranteed to find the optimal policy.

(iii) [*true* or *false*] In Direct Evaluation, you recalculate state values after each transition you experience.

(iv) [*true* or *false*] Q-learning requires that all samples must be from the optimal policy to find optimal q-values.